UX DESIGN & DEVELOPMENT LAB

Web design and development

- HTML fundamentals and page structure
- Lists, tables, hyperlinks, images, forms and CSS
- Online Book Store pages
- PHP programs
- JavaScript validation and interactive programs

UX Design and Development (SDC)

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II YEAR I SEM
CODE:

Course Objectives (COBJs)

- To introduce students to the fundamentals of web design using HTML, CSS, JavaScript, and PHP for creating user-friendly web interfaces.
- To develop practical skills in designing interactive, validated, and responsive web pages following UX principles.
- To enable students to build simple dynamic web applications using client-side and server-side scripting techniques.

Course Outcomes (COs)

At the end of the course, the student will be able to:

- **CO1:** Create structured static web pages using HTML elements such as text, lists, tables, links, images, and forms. *(Understand – L2)*
- **CO2:** Apply CSS techniques to enhance layout, presentation, and usability of web pages. *(Apply – L3)*
- **CO3:** Develop interactive client-side web applications using JavaScript for validation, calculations, and UI behavior. *(Apply – L3)*
- **CO4:** Implement basic server-side functionality using PHP for file handling and logical operations. *(Apply – L3)*

Week 1: a) Develop a web page that displays welcome to HTML.
b) Develop a web page for different headings.
Week 2: a) Develop a web page for paragraph and Line Break
b) Develop a web page for different for text Formatting
Week 3: a) Develop a web page that displays Lists (Ordered & Unordered).
b) Develop a web page for different for table
Week 4: a) Develop a web page that provide Hyperlink.
b) Develop a web page with Image
Week 5: a) Develop a web page that form.
b) Develop a web page for different Outline, Inline CSS
Week 6: a) Develop a web page that Online Book store: the static home page must contain three frames.
b) Develop a web page for the LOGIN PAGE for the Online Book store
Week 7: a) Write a PHP script to print prime numbers between 1-50.
b) Write PHP script to find the length of a string and count no of words in a string.
Week 8: Write a PHP script that reads data from one file and writes into another file.
Week 9: Develop a web page using JavaScript to display the current Date and Time
Week 10: Develop a web page to validate the Registration, user login, user profile and payment by credit card pages using JavaScript.
Week 11: Develop a web page using JavaScript for Even or Odd Number Checker Program
Week 12: Develop a web page using JavaScript for Simple Calculator Program
Week 13: Develop a web page using JavaScript for Form Validation Program
Week 14: Develop a web page using JavaScript for Background Colour Changer

Textbooks

1. **"HTML, CSS, and JavaScript Web Publishing in One"** – Julie C. Meloni
   Covers HTML5, CSS3, and JavaScript basics with practical examples.
2. **"Web Design with HTML, CSS, JavaScript and jQuery Set"** – Jon Duckett
   Well-structured and beginner-friendly for practical web development.

Reference Books

1. **“Learning PHP, MySQL & JavaScript with jQuery, CSS & HTML5”** – Robin Nixon
   Includes PHP scripting, database interaction, and client-side scripting.
2. **“Web Development and Design Foundations with HTML5”** – Terry Felke-Murphy
   Covers UX fundamentals along with hands-on HTML, CSS, and JavaScript.

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	–	2	–	–	–	–	1	–	2
CO2	2	2	3	–	3	–	–	–	–	1	–	2
CO3	2	3	3	1	3	–	–	–	1	2	–	2
CO4	2	2	3	1	3	–	–	–	1	2	–	2

Week 1 a. Basic HTML Page

Program

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Page</title>
</head>
<body>
  <h1>Welcome to HTML</h1>
  <p>This is my first HTML program.</p>
</body>
</html>
```

Output

Displays a heading **"Welcome to HTML"** and a paragraph below it.

Week 1 b. HTML Program with Headings

Program

```
<!DOCTYPE html>
<html>
<body>
  <h1>Main Heading</h1>
  <h2>Sub Heading</h2>
  <h3>Smaller Heading</h3>
</body>
</html>
```

Output

Shows headings from large to smaller size.

Week 2 a. HTML Paragraph and Line Break

Program

```
<!DOCTYPE html>
<html>
<body>
   <p>This is the first line.<br>This is the second line.</p>
</body>
</html>
```

Output

Text appears in two lines.

Week 2 b. HTML Text Formatting

Program

```
<!DOCTYPE html>
<html>
<body>
   <b>Bold Text</b><br>
   <i>Italic Text</i><br>
   <u>Underlined Text</u>
</body>
</html>
```

Output

Text appears in bold, italic, and underlined styles.

Week 3 a. HTML Lists (Ordered & Unordered)

Program

```
<!DOCTYPE html>
<html>
<body>
  <h3>Subjects</h3>
  <ul>
    <li>Maths</li>
    <li>Physics</li>
  </ul>

  <h3>Steps</h3>
  <ol>
    <li>Start</li>
    <li>Process</li>
    <li>End</li>
  </ol>
</body>
</html>
```

Output

Displays bullet points and numbered list.

Week 3 b. HTML Table

Program

```
<!DOCTYPE html>
<html>
<body>
  <table border="1">
    <tr>
      <th>Name</th>
      <th>Marks</th>
    </tr>
    <tr>
      <td>Sania</td>
      <td>90</td>
    </tr>
    <tr>
      <td>Ayesha</td>
      <td>95</td>
    </tr>
  </table>
</body>
</html>
```

Output

Displays a table with names and marks.

Week 4 a. HTML Hyperlink

Program

```
<!DOCTYPE html>
<html>
<body>
   <a href="https://www.google.com">Visit Google</a>
</body>
</html>
```

Output

A clickable link that opens Google.

Week 4 b. HTML Image

Program

```
<!DOCTYPE html>
<html>
<body>
   <img src="https://via.placeholder.com/150" alt="Sample Image">
</body>
</html>
```

Output

Displays a sample image.

Week 5 a. HTML Form

Program

```
<!DOCTYPE html>
<html>
<body>
  <form>
    Name: <input type="text"><br><br>
    Password: <input type="password"><br><br>
    <input type="submit" value="Login">
  </form>
</body>
</html>
```

Output

Shows a form with name, password fields, and a submit button.

Week 5 b. HTML with Inline CSS

Program

```
<!DOCTYPE html>
<html>
<body style="background-color: lightblue;">
  <h1 style="color: red;">Styled Heading</h1>
  <p style="font-size: 18px;">This is a styled paragraph.</p>
</body>
</html>
```

Week 6 (a): Online Book Store – Home Page with Three Frames

1. Main Frames Page (index.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Book Store</title>
</head>

<frameset rows="20%, 80%">

  <!-- Top Frame -->
  <frame src="top.html" name="topFrame">

  <!-- Bottom Frame divided into 2 columns -->
  <frameset cols="30%, 70%">
    <frame src="menu.html" name="menuFrame">
    <frame src="home.html" name="contentFrame">
  </frameset>

</frameset>

</html>
```

2. Top Frame (top.html)

```
<!DOCTYPE html>
<html>
<body style="background-color:lightblue; text-align:center;">
  <h1>📚 Online Book Store</h1>
</body>
</html>
```

3. Menu Frame (menu.html)

```
<!DOCTYPE html>
<html>
<body style="background-color:lightgray;">
  <h3>Menu</h3>
  <ul>
    <li><a href="home.html" target="contentFrame">Home</a></li>
    <li><a href="login.html" target="contentFrame">Login</a></li>
    <li><a href="books.html" target="contentFrame">Books</a></li>
  </ul>
</body>
</html>
```

4. Home Page Content (home.html)

```
<!DOCTYPE html>
<html>
<body>
```

```
  <h2>Welcome to Online Book Store</h2>
  <p>Find your favorite books here!</p>
</body>
</html>
```

5. Books Page (books.html)

```
<!DOCTYPE html>
<html>
<body>
  <h2>Available Books</h2>
  <ul>
    <li>Java Programming</li>
    <li>Python Basics</li>
    <li>Data Structures</li>
    <li>Web Development</li>
  </ul>
</body>
</html>
```

Week 6 (b): Login Page for Online Book Store

login.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Login Page</title>
</head>
<body>

  <h2>Login to Book Store</h2>

  <form onsubmit="return validateLogin()">
    Username: <input type="text" id="username"><br><br>
    Password: <input type="password" id="password"><br><br>

    <input type="submit" value="Login">
  </form>

  <p id="error" style="color:red;"></p>

  <script>
    function validateLogin() {
      let user = document.getElementById("username").value;
      let pass = document.getElementById("password").value;

      if (user === "" || pass === "") {
        document.getElementById("error").innerHTML = "All fields are required";
```

```
            return false;
        }

        // Simple validation (demo purpose)
        if (user === "admin" && pass === "1234") {
            alert("Login Successful!");
            return true;
        } else {
            document.getElementById("error").innerHTML = "Invalid Username or Password";
            return false;
        }
    }
  </script>

</body>
</html>
```

Week 7 a. Write a PHP script to print prime numbers between 1-50.

```
<?php
$number = 2 ;
while ($number <100 )
{
$div_count=0;
for ( $i=1;$i<=$number;$i++)
{
if (($number%$i)==0)
{
$div_count++;
}
}
if ($div_count<3)
{
echo $number." , ";
}
$number=$number+1;
}
    ?>
```

OUTPUT:2 , 3 , 5 , 7 , 11 , 13 , 17 , 19 , 23 , 29 , 31 , 37 , 41 , 43 , 47

Week 7 b PHP script to Find the length of a string.

```
<!DOCTYPE html>
<html>
<body>
<?php
echo strlen("Hello world!");
?>

</body>
</html>
```

OUTPUT:

12

Count no of words in a string.

```
<!DOCTYPE html>
<html>
<body>

<?php
echo str_word_count("Hello world!");
?>
```

```
</body>
</html>
```

OUTPUT:

2

Week 8: Write a PHP script that reads data from one file and write into another file.

```
<?php
if(isset($_POST['save']))

{

$f=$_POST['file'];

$ext=$_POST['ext'];

$data=$_POST['data'];

$file=$f.$ext;

if(file_exists($file))

{

echo "<font color='red'>file already exists</font>";

}

else

{

$fo = fopen($file,"w");

fwrite($fo,$data);

echo "your data is saved";

}

    }
```

```
?>

<form method="post">

enter your file<input type="text" name="file"/><br/>

choose your extension<select name="ext">
<option value="">choose urexten</option>
<option>.txt</option>
<option>.docs</option>
<option>.pdf</option>
</select><br/>

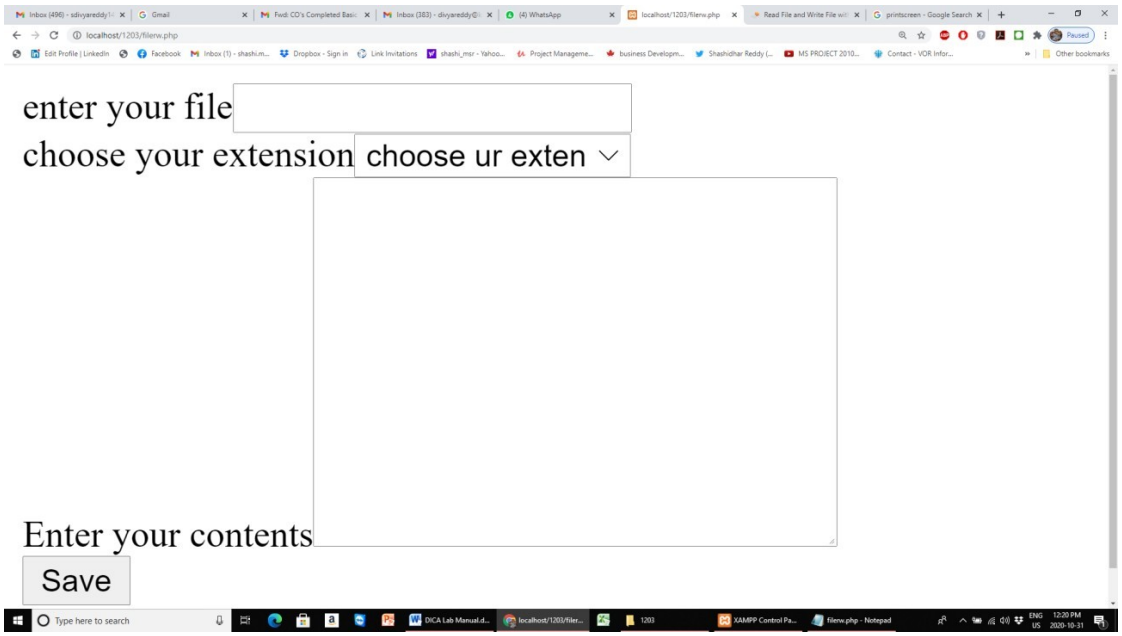
Enter your contents<textarea rows="10" cols="30" name="data">

<?php echo @$contents ; ?>

</textarea><br/>

<input type="submit" value="Save" name="save"/>

    </form>
```




```
<?php

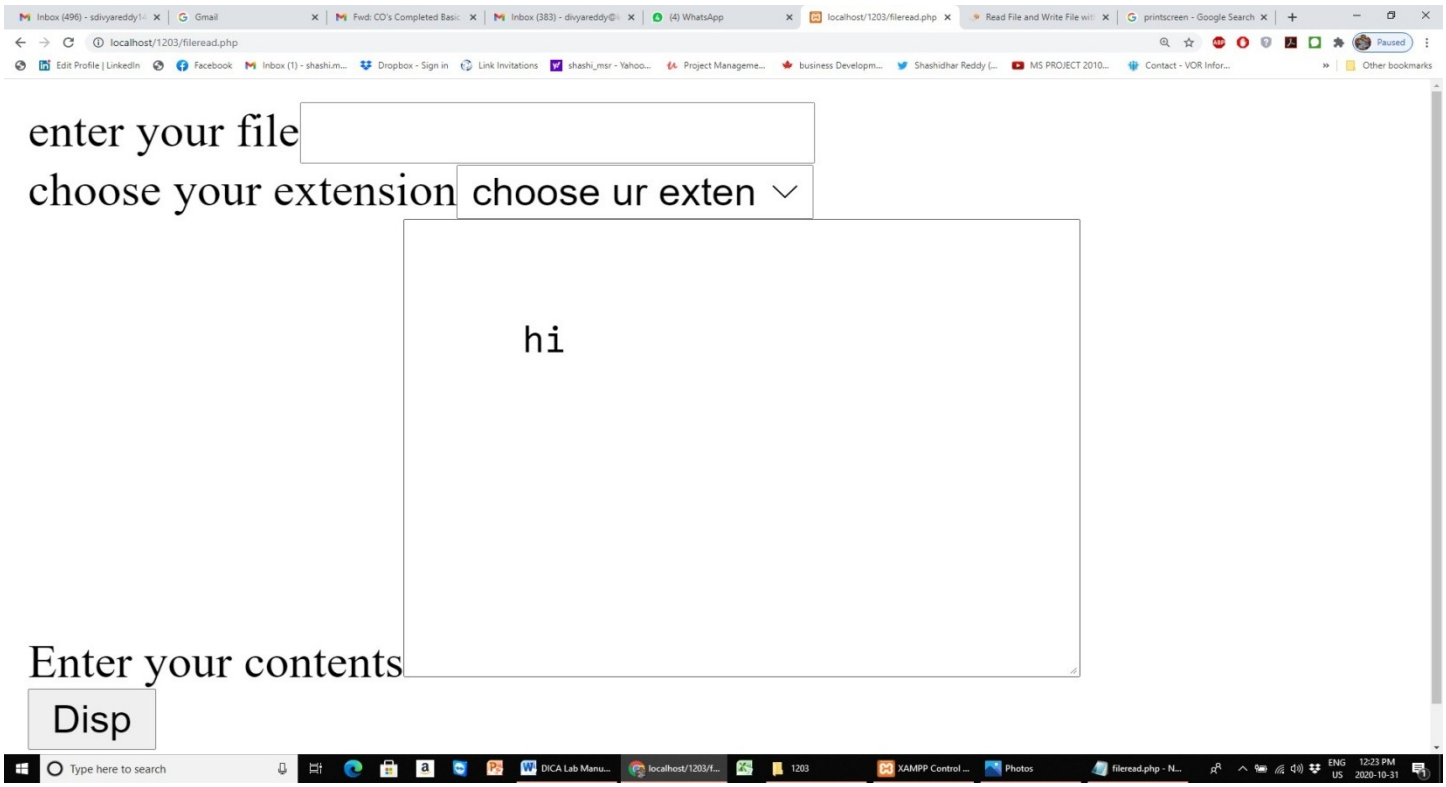
if(isset($_POST['disp']))
```

```
{
$f=$_POST['file'];
$ext=$_POST['ext'];
$file=$f.$ext;
if(file_exists($file))
{
$fo = fopen($file,"r");
$contents = fread($fo,filesize($file));
}
else
{
echo "<font color='red'>file doesn't  exists</font>";
}
}
?>
<form method="post">
enter your file<input type="text" name="file"/><br/>
choose your extension<select name="ext">
<option value="">choose urexten</option>
<option>.txt</option>
<option>.docs</option>
<option>.pdf</option>
</select><br/>
Enter your contents<textarea rows="10" cols="30" name="data">
<?php echo @$contents ; ?>
```

```
</textarea><br/>

<input type="submit" value="Disp" name="disp"/>

</form>
```



1. **Validate the Registration, user login, user profile and payment by credit card pages using JavaScript.**

- **Homepage:**

Main.html:

```
<html>

<frameset rows="25%,*">
<frame src="top.html" name="top" scrolling ="no" frameborder ="0">
<frameset cols="25%,75%">
<frame src="left.html" name="left" scrolling ="no" frameborder ="0">
<frame src="right.html" name="right" scrolling ="auto" frameborder ="0">
</frameset>
</frameset>
</html>
```

Top.html:

```
<html>
<body bgcolor="pink">
<br><br>
<marquee><h1
align=”center”><b><u>ONLINE BOOK
STORAGE</u></b></h1></marquee>
</body>
</html>
```

Right.html:

```
<html>
<body>
<br><br><br><br><br>
<h2 align="center">
<b><p> welcome to online book storage. Press
login if you are having id otherwise press
registration.
</p></b></h2>
</body></html>
```

Left.html:

```
<html>
  <body bgcolor="pink">
  <h3>
  <ul>
  <li><a href="login.html"
  target="right"><font color="black">
  LOGIN</font></a></li><br><br>
  <li><a href="profile.html" target="right"><font
  color="black"> USER
  PROFILE</font></a></li><br><br>
  <li><a href="catalog.html"
  target="right"><font color="black"> BOOKS
  CATALOG</font></a></li><br><br>
  <li><a href="scart.html" target="right"><font
  color="black">
  SHOPPINGCART</font></a></li><br><br>
  <li><a href="payment.html" target="right"><font
  color="black">
  PAYMENT</font></a></li><br><br>
  <br><br>
  </ul>
  </body>
  </html>
```

Registration and user Login

Login.html:

```
<html>
<body bgcolor="pink"><br><br><br>
<script
language="javascript">
function validate()
{
var flag=1;
if(document.myform.id.value==""
"||
document.myform.pwd.value==
"")
{
alert("LoginId and Password must be filled")
flag=0;
}
if(flag==1)
{
```

```
alert("VALID INPUT");
window.open("catalog.html","right");
}
else
{
alert("INVALID INPUT");
//document.myform.focus();
}
}
</script>
<form name="myform">
<div align="center"><pre>
LOGIN ID:<input type="text" name="id"><br>
PASSWORD:<input type="password"
name="pwd"><br><br>
</pre>
<input type="button" value="ok" onClick="validate()">    
<input type="reset" value="clear" >
</div>

</form>
</body>
</html>
```

User profilepage

Profile.html:

```
<html>
        <body bgcolor="pink"><br><br>
        <script
         type="text/javasc
         ript"> function
         validate()
          {
            var flag=1;
             if(document.myform.name.value
             ==""||
                document.myform.add
                r.value==""||
                document.myform.phn
                o.value==""||
                document.myform.id.v
                alue==""||
                document.myform.pw
```

```
d.value=="")
 {
                alert("Enter all the details");
   flag=0;
 }
  var str=document.myform.phno.value;
                var x=new RegExp("\\d","g");
                if(!(str.match(x)))
                {
                        if(!(str.
                        length
                        ==10))
                        flag=0
                        ;
                }
                var str1=document.myform.id.value;
                var x1=new RegExp("^[A-Z][a-
                zA-Z]+$","g");
                if(!(str1.match(x1)))
                {
                        flag=0;
                        alert("Invalid UserID");
                        }
                        var
                str1=document.myform.pwd.valu
                e; var x1=new RegExp("^[A- Z]
                [a-zA-Z]+$","g");
                if(!(str1.match(x1)))
                {
                  flag=0;
                  alert("Invalid password");
                  }
 if(f
 lag
 ==
 1)
      {
       alert("VALID INPUT");
    window.self.location.href="login.html";
         }
```

```
                                    else
                                   {
                                        alert("INVALIDINPUT");
                                        document.myform.focus();
<br><br>
}
}
</script>
        <form name="myform">
         <div align="center"><pre>
          NAME :<input type="text" name="name"><br> ADDRESS :<input
          type="type" name="addr"><br> CONTACT NUMBER:<input
          type="text"name="phno"><br> LOGINID :<input
          type="text"name="id"><br>
          PASSWORD :<input type="password"name="pwd"></pre><br><br>
         </div>

<div align="center">
<input type="button" value="ok" onClick="validate()">   
<input type="reset" value="clear">
              </form></body></html>
```

- **Bookscatalog:**

Scart.html:

```
<html>
<body bgcolor="pink"><br><br><br>
             <script
                language="javascript
                "> function
                validate()
                 {
                   var flag=1;
                    if(document.myform.title.val
                    ue=="")
                       {
                         flag=0;
                       }

                           str=document.myform.title.value;

                           if(str=="c"||str=="C")
```

```
                    {
                     document.myform.t1.value="C";
                     document.myform.t2.value=444;
                     }
                else if(str=="jsp"||str=="JSP")




else{
 document.myform.t1.value="JSP";  document.myform.t2.value=555;
 }

{
 flag=0;
}
```

- ***Shoppingcart:***

Catalog.html:

```
<html>
<body bgcolor="pink"><br><br><br>
        <script
          language="javascript
          "> function
          validate()
           {
            var flag=1;
             if(document.myform.id.value
             ==""||
               document.myform.title.valu
               e==""||
               document.myform.no.valu
               e==""||
               document.myform.cost.val
               ue=="")
                {
                  flag=0;
               }


             str=document.myform.title.value;
             var str1=document.myform.cost.value;
             if(!((str=="c"&& str1==444) || (str=="jsp" && str1==555)))
                        {
```

```
                                flag=0;
                                }
if(flag==1)
{

alert("VALID INPUT");


else
}

                    }

   {
     alert("INVALIDINPUT");
     document.myform.focus();
       }
       </script>
<form name="myform" action="scart.html" target="right">
<div align="center"><pre>
LOGINID              :<input type="text"
name="id"><br> TITLE                          :<input
type="text" name="title"><br>NO.OFBOOKS
                    :<input type="text"name="no"><br>
COSTOFBOOK              :<input type="text"name="cost"><br>
</pre><br><br>
</div>
<br><br>
<div align="center">
<input type="submit" value="ok" onClick="validate()">    
<input type="reset" value="clear">
</form>
</body>
</html>
```

- **Payment by creditcard**

Payment.html:

```
<html>
<body bgcolor="pink"><br><br><br>
<script language="javascript">
function validate()
{
var flag=1;
if(document.myform.id.value==""|| document.myform.pwd.value==""||
document.myform.amount.value==""|| document.myform.num.value=="")
{
        flag=0;
}
var str=document.myform.amount.value;
var x=new RegExp("\\d","g");
if(!(str.match(x)))
{
{
flag=0;
}
if(flag==1)
{
alert("VALID INPUT");
window.self.location.href="order.html";
}
else
{
alert("INVALIDINPUT");
document.myform.focus();
}
}
</script>
<form name="myform">
<div align="center"><pre>
LOGINID                 :<input type="text" name="id"><br>
PASSWORD                :<input type="password" name="pwd"><br>
AMOUNT                          :<input
type="text"name="amount"><br>
CREDITCARDNUMBER        :<input type="PASSWORD"name="num"><br></pre><br><br>
</div>
<br><br>
<div align="center">
<input type="button" value="ok" onClick="validate()">    
<input type="reset" value="clear" >
</form>
</body>
```

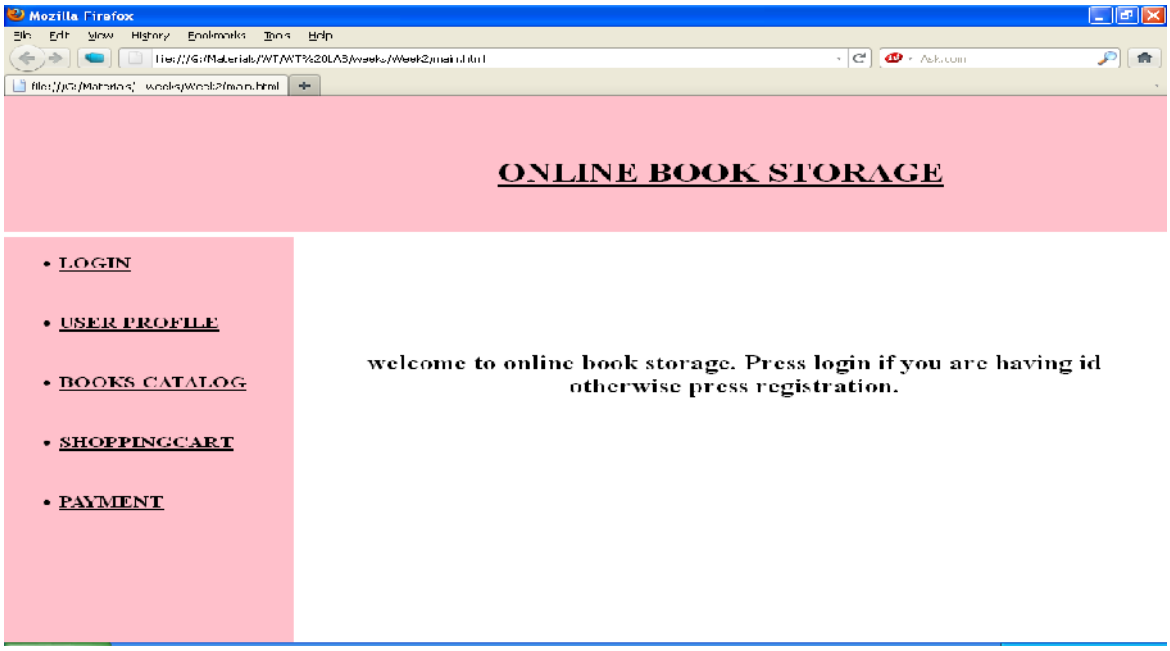
```
</html>
```

- **Order ConformationOrder.html:**

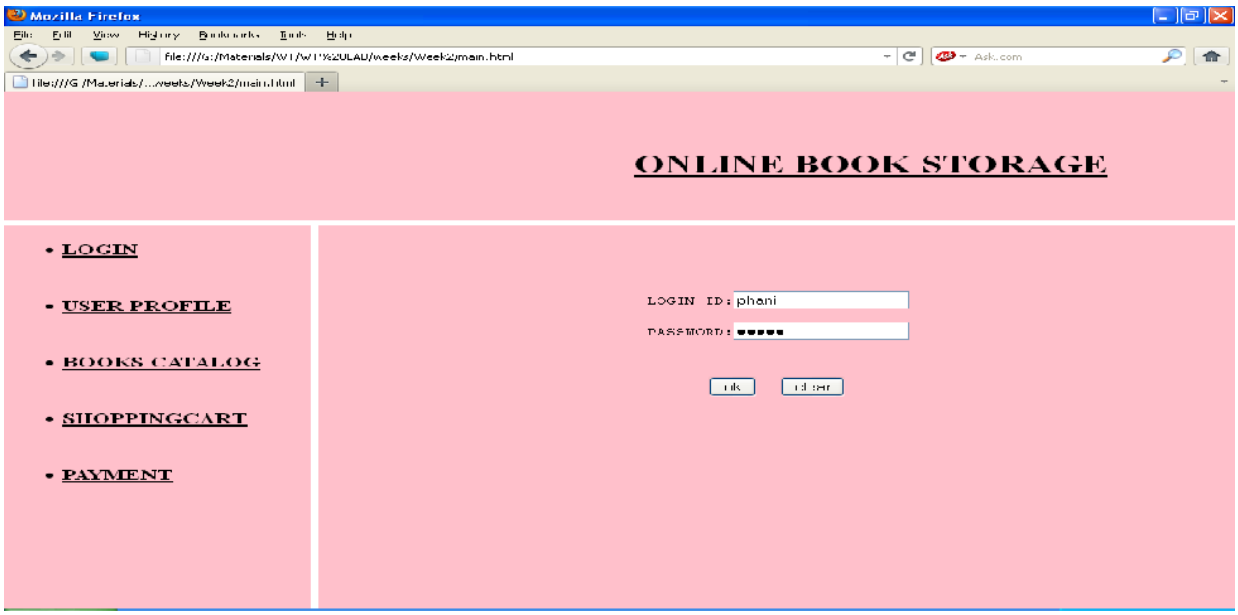
```
<html>
<head><title>order conformation</title><M/head>
<body bgcolor="cyan">
<center>
<h1><b>AMAZON</h1>
<pre><strong>
<b>Your order Is Conformed
</strong></pre>
<h2><b>THANK YOU</h2>
</center>
</body>
</html>
```

OUTPUT:

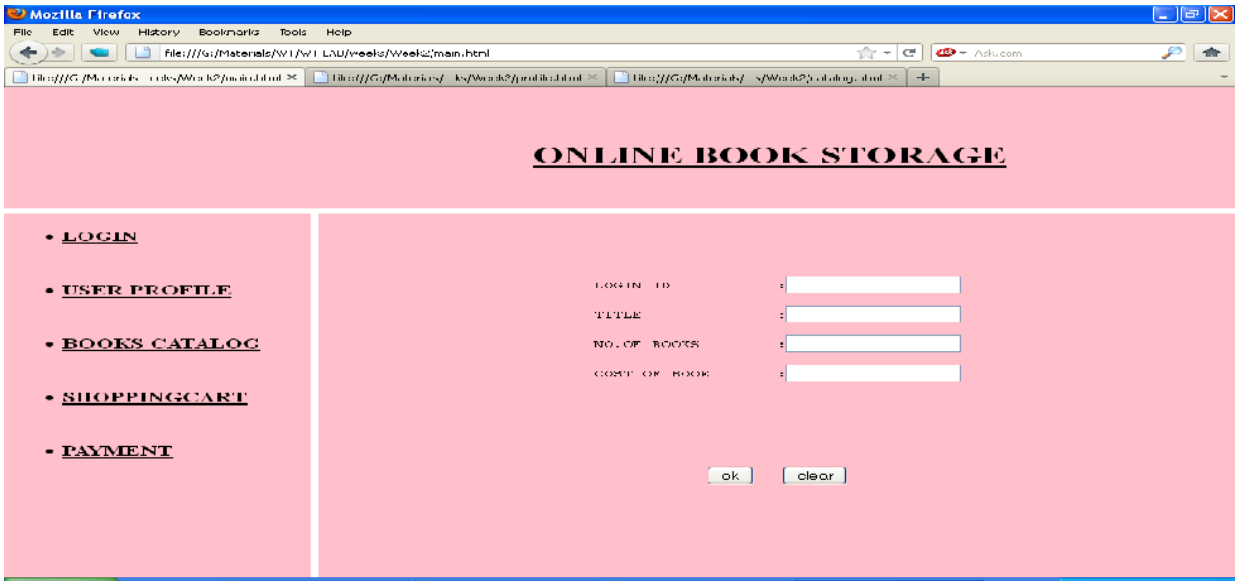
Main.html



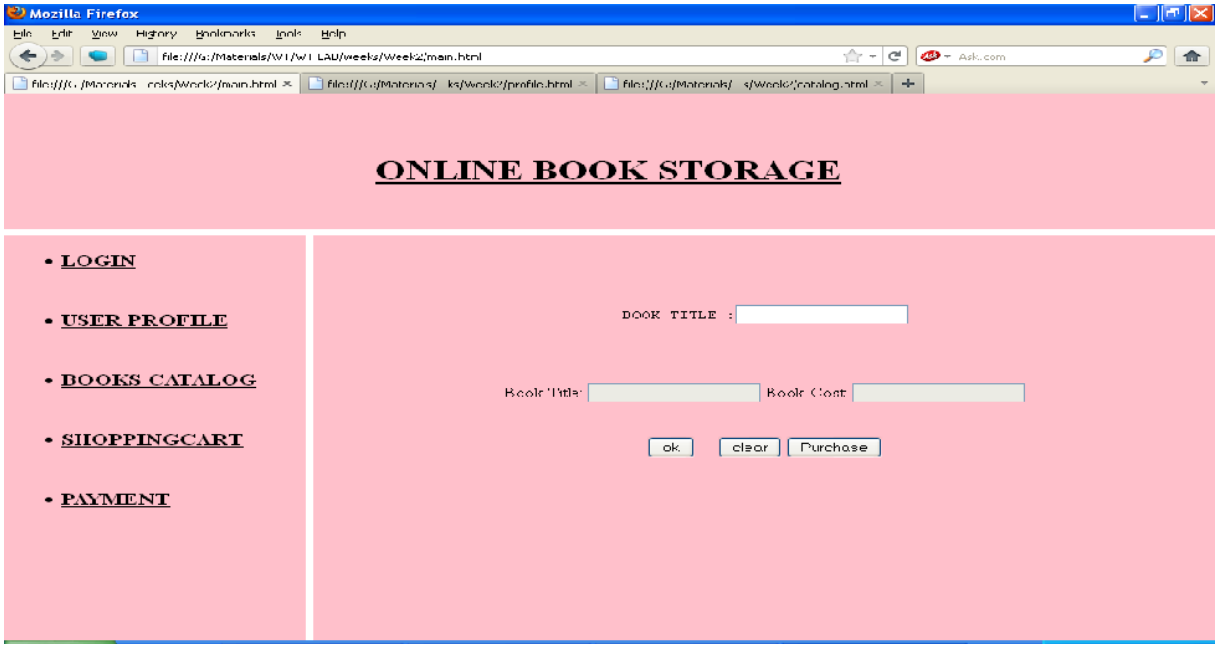
Login.html:



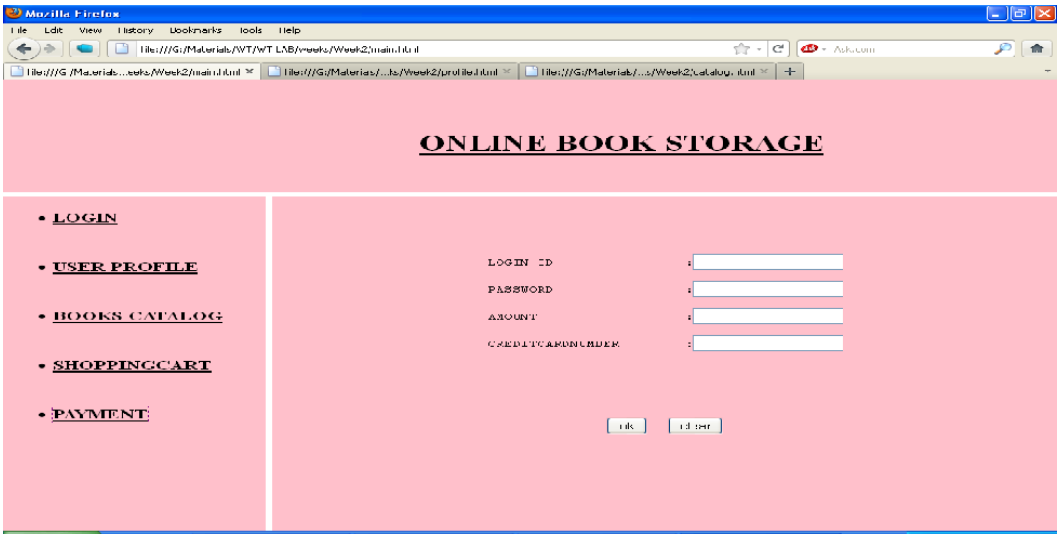
Catalog.html:



Scart.html:


Mozilla Firefox
ONLINE BOOK STORAGE
LOGIN
USER PROFILE
BOOKS CATALOG
SHOPPINGCART
PAYMENT
BOOK TITLE :
Book Title:
Book Cost:
ok
clear
Purchase

Payment.html:



Order.html



Week 11: Even or Odd Number Checker

```
<!DOCTYPE html>
<html>
<head>
   <title>Even or Odd Checker</title>
</head>
<body>
   <h2>Even or Odd Number Checker</h2>

   <input type="number" id="num" placeholder="Enter a number">
   <button onclick="check()">Check</button>

   <p id="result"></p>

   <script>
      function check() {
         let num = document.getElementById("num").value;

         if (num === "") {
            document.getElementById("result").innerHTML = "Please enter a number";
            return;
         }

         if (num % 2 == 0) {
            document.getElementById("result").innerHTML = num + " is Even";
         } else {
            document.getElementById("result").innerHTML = num + " is Odd";
         }
      }
   </script>
</body>
</html>
```

Output: 5
5 is Odd

Week 12: Simple Calculator

```
<!DOCTYPE html>
<html>
<head>
   <title>Simple Calculator</title>
</head>
<body>
   <h2>Simple Calculator</h2>

   <input type="number" id="num1" placeholder="Enter first number">
   <input type="number" id="num2" placeholder="Enter second number"><br><br>

   <button onclick="calculate('+')">Add</button>
   <button onclick="calculate('-')">Subtract</button>
   <button onclick="calculate('*')">Multiply</button>
   <button onclick="calculate('/')">Divide</button>

   <p id="result"></p>

   <script>
      function calculate(op) {
         let n1 = parseFloat(document.getElementById("num1").value);
         let n2 = parseFloat(document.getElementById("num2").value);
         let result;

         if (isNaN(n1) || isNaN(n2)) {
            document.getElementById("result").innerHTML = "Enter valid numbers";
            return;
         }

         switch(op) {
            case '+': result = n1 + n2; break;
            case '-': result = n1 - n2; break;
            case '*': result = n1 * n2; break;
            case '/':
               if(n2 == 0) {
                  result = "Cannot divide by zero";
               } else {
                  result = n1 / n2;
               }
               break;
         }

         document.getElementById("result").innerHTML = "Result: " + result;
      }
   </script>
</body>
</html>
```

Output: 2 3 result 5

Week 13: Form Validation Program

```
<!DOCTYPE html>
<html>
<head>
  <title>Form Validation</title>
</head>
<body>
  <h2>Form Validation</h2>

  <form onsubmit="return validate()">
    Name: <input type="text" id="name"><br><br>
    Email: <input type="text" id="email"><br><br>
    Password: <input type="password" id="password"><br><br>

    <input type="submit" value="Submit">
  </form>

  <p id="error" style="color:red;"></p>

  <script>
    function validate() {
      let name = document.getElementById("name").value;
      let email = document.getElementById("email").value;
      let password = document.getElementById("password").value;

      if (name == "") {
        document.getElementById("error").innerHTML = "Name is required";
        return false;
      }

      if (!email.includes("@") || !email.includes(".")) {
        document.getElementById("error").innerHTML = "Enter valid email";
        return false;
      }

      if (password.length < 6) {
        document.getElementById("error").innerHTML = "Password must be at least 6 characters";
        return false;
      }

      alert("Form submitted successfully!");
      return true;
    }
  </script>
</body>
</html>
```

Output: Form submitted successfully!

Week 14: Background Colour Changer

```
<!DOCTYPE html>
<html>
<head>
   <title>Background Color Changer</title>
</head>
<body>
   <h2>Background Colour Changer</h2>

   <button onclick="changeColor('red')">Red</button>
   <button onclick="changeColor('blue')">Blue</button>
   <button onclick="changeColor('green')">Green</button>
   <button onclick="changeColor('yellow')">Yellow</button>
   <button onclick="randomColor()">Random</button>

   <script>
      function changeColor(color) {
         document.body.style.backgroundColor = color;
      }

      function randomColor() {
         let colors = ["red", "blue", "green", "yellow", "pink", "orange"];
         let random = colors[Math.floor(Math.random() * colors.length)];
         document.body.style.backgroundColor = random;
      }
   </script>
</body>
</html>
```

Output: